22 forecasters on the same test windows of nine populations. a, b: means over units, so larger units weigh more; c, d: the median unit. Bold: lowest in the column. Rows in the order of figure 2. ankyra-lite 0.2.0: carrier recipe Rw (trained from scratch with early stopping on the 93-day validation set, then 8 more epochs with the validation span returned and each training window's loss weighted by the inverse-noise rule; six seeds averaged; figure 1b), the anchoring layer, the state rule and the origin-continuity layer; hourly errors score the delivered trajectory after that layer, energy, level and peak are read before it (readout rule A). Linear-context: the TiDE implementation of the release, whose forecast is a linear map of the load context.

a RMSE (kW, mean over units)

	BDG2	Cambridge	HEEW	EWELD	GoiEner NH	GoiEner HH	Oslo	Drammen	CINELDI
ankyra-lite	**28.8**	**12.7**	26.1	98.8	**0.550**	**0.170**	**22.5**	**13.3**	**16.3**
Ridge (per unit)	30.1	15.0	29.3	103	0.590	0.173	28.9	14.7	20.6
Chronos-2-X	30.1	14.5	27.3	106	0.562	0.177	26.2	14.4	19.0
iTransformer-X	32.9	13.8	31.1	103	0.641	0.180	25.2	14.9	18.2
iTransformer	33.2	14.5	30.8	105	0.633	0.183	26.9	15.3	17.2
PatchTST	32.8	14.7	28.3	104	0.651	0.178	26.5	15.2	17.2
TimesFM-X	32.0	15.8	27.7	102	0.606	0.176	29.0	15.8	21.1
GBT-T	30.6	17.6	**25.5**	154	0.809	0.183	23.1	14.8	16.7
TimesFM	31.2	15.6	27.6	100	0.643	0.177	29.1	17.0	21.4
Chronos-2	31.9	15.9	29.6	**97.1**	0.637	0.177	29.1	16.4	18.5
DLinear	30.7	16.8	32.2	121	0.640	0.180	29.5	15.4	18.6
Linear-context	30.7	17.4	32.3	122	0.644	0.180	29.6	15.4	18.4
4-week profile	34.8	15.2	38.2	111	0.729	0.188	31.6	18.9	18.8
LSTM-T-KAN-M	30.8	29.4	38.0	160	0.687	0.184	38.1	17.5	19.8
Holt–Winters	37.0	16.9	33.1	115	0.723	0.196	33.6	18.4	27.1
LSTM	32.9	27.9	39.7	144	0.692	0.182	39.4	18.3	22.6
Previous-month profile	36.9	16.7	39.9	110	0.775	0.185	36.4	20.4	21.6
MSTL	35.8	17.8	30.6	109	0.731	0.213	34.9	18.7	26.0
Seasonal naive (week)	36.2	17.9	33.2	115	0.727	0.214	34.5	18.6	24.0
Last-year profile	37.6	21.6	38.8	249	0.698	0.246	33.1	18.8	21.5
GBT (zero-shot)	52.0	18.2	41.2	135	0.846	0.231	44.3	20.5	24.9
Seasonal naive (day)	40.9	22.6	36.1	106	0.826	0.216	47.0	24.8	31.9

b MAE (kW, mean over units)

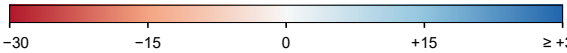
	BDG2	Cambridge	HEEW	EWELD	GoiEner NH	GoiEner HH	Oslo	Drammen	CINELDI
ankyra-lite	20.4	**7.80**	19.3	66.0	0.343	0.100	**15.2**	**8.96**	11.5
Ridge (per unit)	21.5	9.33	21.0	66.9	0.362	0.104	19.9	10.1	13.7
Chronos-2-X	**20.4**	8.76	19.1	67.2	**0.311**	**0.0927**	16.7	9.62	10.5
iTransformer-X	23.5	8.56	22.2	69.7	0.403	0.109	16.8	10.2	12.5
iTransformer	24.0	9.01	22.1	70.1	0.398	0.110	17.8	10.3	11.9
PatchTST	23.8	9.22	20.7	70.7	0.418	0.106	17.9	10.5	12.5
TimesFM-X	22.6	9.76	19.3	65.1	0.358	0.0963	19.8	11.0	13.3
GBT-T	21.9	9.96	**18.4**	90.8	0.489	0.114	15.7	10.2	11.4
TimesFM	21.4	9.35	19.0	63.4	0.362	0.0943	18.9	11.3	12.7
Chronos-2	21.9	9.56	20.2	**60.7**	0.353	0.0927	18.2	10.7	**10.0**
DLinear	22.4	11.3	24.6	93.7	0.418	0.112	20.4	10.5	12.8
Linear-context	22.4	11.9	24.6	94.8	0.425	0.111	20.6	10.6	12.6
4-week profile	25.1	9.19	27.3	72.8	0.430	0.109	21.0	13.5	11.6
LSTM-T-KAN-M	22.4	22.8	28.8	127	0.455	0.114	27.9	11.8	15.0
Holt–Winters	26.8	10.7	24.2	78.4	0.447	0.126	22.4	12.9	18.8
LSTM	24.6	21.8	30.5	118	0.467	0.115	29.3	12.6	17.6
Previous-month profile	28.2	11.5	29.5	73.9	0.521	0.112	28.1	15.3	16.9
MSTL	25.2	11.2	21.6	71.0	0.449	0.135	23.2	12.6	17.1
Seasonal naive (week)	24.4	11.0	22.3	72.5	0.399	0.117	21.7	12.4	13.9
Last-year profile	26.0	14.5	28.5	196	0.409	0.139	22.0	12.9	13.0
GBT (zero-shot)	30.8	11.8	27.5	84.9	0.519	0.139	27.8	14.0	17.9
Seasonal naive (day)	27.9	14.5	24.9	66.6	0.477	0.122	30.3	16.3	20.9

c CV(RMSE) (%, median unit)

	BDG2	Cambridge	HEEW	EWELD	GoiEner NH	GoiEner HH	Oslo	Drammen	CINELDI
ankyra-lite	19.7	**16.1**	**16.3**	53.9	**69.1**	**96.9**	**28.2**	**29.3**	**38.6**
Ridge (per unit)	22.6	19.4	17.9	55.3	71.1	98.6	36.1	32.7	43.2
Chronos-2-X	**18.1**	18.8	17.0	52.5	78.0	102.6	31.8	32.2	44.4
iTransformer-X	23.2	17.6	18.3	55.1	74.8	101.0	31.9	32.8	47.3
iTransformer	22.9	19.5	19.3	57.7	76.2	103.0	33.0	32.4	45.0
PatchTST	23.4	19.3	19.8	55.8	78.8	101.9	33.0	33.2	42.9
TimesFM-X	23.0	20.9	18.1	56.3	78.0	101.7	37.5	35.4	44.0
GBT-T	19.5	17.7	18.6	68.6	89.3	102.9	30.4	33.6	46.4
TimesFM	20.9	18.7	17.5	55.9	77.4	102.9	37.5	37.1	43.7
Chronos-2	21.9	20.4	18.5	**51.9**	76.8	102.4	36.8	35.5	41.5
DLinear	23.0	22.7	19.7	56.8	80.3	102.1	37.1	34.9	43.7
Linear-context	23.3	23.9	19.8	56.7	81.0	102.0	37.7	35.1	44.0
4-week profile	24.5	19.9	21.0	56.5	83.5	106.9	41.3	42.3	48.3
LSTM-T-KAN-M	23.1	32.7	24.6	64.1	83.2	103.1	43.8	39.5	46.0
Holt–Winters	25.1	21.1	21.1	68.7	83.8	110.4	44.6	41.6	57.5
LSTM	25.0	29.3	25.3	65.8	85.0	102.4	42.9	42.8	56.1
Previous-month profile	25.8	23.5	22.0	57.4	82.3	104.6	45.9	48.5	52.5
MSTL	26.0	23.0	20.0	67.1	90.7	123.0	44.2	42.0	53.9
Seasonal naive (week)	25.1	23.3	20.5	67.1	93.8	125.0	43.5	42.8	51.1
Last-year profile	25.8	27.9	27.2	94.0	97.8	133.4	40.4	43.2	64.0
GBT (zero-shot)	34.4	24.4	25.5	74.1	94.1	120.8	48.3	48.3	58.7
Seasonal naive (day)	29.1	30.0	21.6	63.7	102.9	124.0	59.4	60.0	77.1

d WAPE (%, median unit)

	BDG2	Cambridge	HEEW	EWELD	GoiEner NH	GoiEner HH	Oslo	Drammen	CINELDI
ankyra-lite	13.9	**11.3**	11.6	34.1	40.7	56.5	**19.1**	**19.6**	25.8
Ridge (per unit)	15.8	13.1	12.2	34.1	43.6	58.8	25.7	22.6	28.2
Chronos-2-X	**12.6**	11.8	**10.8**	32.6	39.9	52.3	20.5	21.3	**25.3**
iTransformer-X	16.7	12.7	13.3	35.5	45.4	60.4	20.6	22.8	33.0
iTransformer	17.1	13.7	13.3	36.9	44.3	61.0	21.8	21.6	28.7
PatchTST	17.2	13.6	13.9	38.3	48.9	59.5	21.6	22.9	29.3
TimesFM-X	15.8	12.9	12.2	33.6	43.2	54.4	25.2	25.3	28.2
GBT-T	14.6	12.8	13.0	42.4	50.7	61.5	20.7	22.8	31.0
TimesFM	14.4	12.3	11.3	32.7	40.5	54.0	23.9	25.2	27.4
Chronos-2	15.5	12.6	11.6	**31.2**	**39.7**	**52.0**	24.0	23.1	26.6
DLinear	16.6	16.4	14.7	41.2	49.7	62.3	25.8	23.6	32.3
Linear-context	16.6	17.2	14.6	41.9	50.7	62.2	25.9	23.8	32.9
4-week profile	17.8	12.9	14.0	36.7	44.9	61.8	28.4	30.4	31.6
LSTM-T-KAN-M	17.2	23.0	18.1	46.8	52.2	62.7	30.2	27.1	35.0
Holt–Winters	18.8	14.8	14.5	43.7	50.8	65.9	28.9	28.4	37.8
LSTM	18.5	18.9	18.3	49.4	52.5	63.0	30.7	29.0	41.7
Previous-month profile	20.5	16.7	16.4	38.0	54.3	62.9	35.0	36.4	39.3
MSTL	17.6	15.1	14.0	40.5	53.9	77.2	29.3	28.1	38.0
Seasonal naive (week)	16.4	14.5	13.6	38.0	48.6	68.6	28.1	27.9	31.4
Last-year profile	17.3	19.2	20.3	70.5	55.7	75.3	26.5	29.0	42.0
GBT (zero-shot)	22.2	16.2	17.2	44.8	57.9	72.3	34.0	32.6	40.2
Seasonal naive (day)	20.1	19.3	15.3	37.6	54.0	70.5	38.0	40.0	49.1



Loss relative to ankyra-lite in the same column (%): red lower than ankyra-lite, blue higher

Same information · trained
Same information · zero-shot foundation model
Load and calendar · trained
Load only · zero-shot foundation model
Load only · statistical
Load only · trained
Per-unit regression (calendar, temperature)
Profiles, naive, zero-shot GBT